

VORKUTA, Russian Federation

WMO: 232260

Lat: 67.4919N

Lon: 63.9754E

Elev: 564

StdP: 14.40

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.2	-33.5	-44.5	0.4	-39.0	-38.7	0.6	-33.3	38.7	9.8	34.9	10.1	4.6	180	0.925

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	77.8	63.1	72.9	60.6	68.2	58.3	64.8	74.1	62.3	71.0	59.6	66.7	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	83.2	68.5	58.5	74.8	66.5	55.7	67.5	63.6	30.2	73.6	28.3	71.4	26.4	66.5	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.3	27.5	23.8	DB	-41.8	82.6	6.7	3.8	-46.6	85.3	-50.5	87.6	-54.3	89.7	-59.2	92.5	(4)
(5)			WB	-41.5	69.2	6.1	4.2	-45.9	72.3	-49.5	74.7	-52.9	77.1	-57.4	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	-2.5	-3.1	6.8	16.6	29.0	46.4	55.2	49.3	40.1	26.4	9.5	3.0	(6)
(7)		DBStd	23.54	15.69	16.05	14.27	12.46	9.96	10.34	9.45	6.82	6.77	9.93	14.31	16.72	(7)
(8)		HDD50	10167	1629	1488	1338	1001	655	192	52	101	307	731	1216	1457	(8)
(9)		HDD65	15286	2094	1908	1803	1451	1117	564	328	489	748	1196	1666	1922	(9)
(10)		CDD50	386	0	0	0	0	3	82	214	78	9	0	0	0	(10)
(11)		CDD65	31	0	0	0	0	0	4	26	1	0	0	0	0	(11)
(12)		CDH74	315	0	0	0	0	0	48	258	9	0	0	0	0	(12)
(13)		CDH80	54	0	0	0	0	0	7	47	0	0	0	0	0	(13)
(14)	Wind	WSAvg	11.4	11.9	11.9	12.2	12.8	11.8	11.1	9.9	10.0	10.2	11.0	11.1	12.6	(14)
(15)	Precipitation	PrecAvg	21.3	1.3	1.3	1.2	1.1	1.5	2.0	2.3	2.7	2.2	2.4	1.6	1.5	(15)
(16)		PrecMax	26.6	2.3	2.5	3.4	2.3	2.6	4.0	4.0	4.3	3.9	4.2	2.9	2.9	(16)
(17)		PrecMin	14.4	0.5	0.3	0.3	0.3	0.7	0.8	0.7	0.7	0.1	1.1	0.9	0.7	(17)
(18)		PrecStd	2.8	0.4	0.6	0.6	0.5	0.5	0.8	1.0	1.1	1.1	0.9	0.4	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	29.2	34.4	42.1	64.2	79.4	84.6	74.7	64.4	49.5	34.2	32.3	(19)
(20)			MCWB	29.1	28.3	31.7	37.9	52.0	60.6	65.7	64.7	55.3	46.4	32.9	31.5	(20)
(21)		2%	DB	25.2	26.2	31.5	38.2	54.4	73.8	80.6	68.9	58.2	44.3	32.3	29.5	(21)
(22)			MCWB	24.3	25.4	30.1	35.1	47.2	58.9	64.5	59.5	51.7	42.4	31.6	28.6	(22)
(23)		5%	DB	21.9	23.1	29.2	35.6	47.8	68.9	76.5	65.0	54.2	41.0	30.7	27.0	(23)
(24)			MCWB	21.1	22.2	27.8	33.3	42.2	56.9	63.0	57.7	49.6	39.4	30.1	25.9	(24)
(25)		10%	DB	17.8	18.8	26.3	33.3	42.5	63.9	71.9	61.4	50.6	38.1	27.7	24.3	(25)
(26)			MCWB	16.7	17.8	25.0	31.8	39.0	54.1	60.6	56.2	47.3	36.8	26.8	23.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.3	28.5	32.3	38.5	53.2	64.0	69.0	67.2	57.0	46.5	33.3	31.7	(27)
(28)			MCD B	29.8	28.9	33.9	41.4	61.7	73.2	78.9	72.1	61.0	48.6	33.7	32.1	(28)
(29)		2%	WB	24.3	25.5	30.2	35.4	47.2	60.4	66.5	61.4	53.1	42.7	31.8	28.7	(29)
(30)			MCD B	25.0	26.0	31.3	37.6	54.1	70.1	76.8	66.4	57.0	44.1	32.2	29.3	(30)
(31)		5%	WB	21.1	22.3	27.9	33.5	42.5	57.5	64.1	58.8	49.9	39.6	29.9	26.1	(31)
(32)			MCD B	21.8	23.2	29.2	35.2	46.7	67.4	73.8	63.7	53.0	40.6	30.7	26.9	(32)
(33)		10%	WB	16.8	18.0	25.1	31.9	38.9	54.7	62.0	56.2	47.5	36.9	26.8	23.4	(33)
(34)			MCD B	17.7	18.9	26.3	33.4	42.6	63.2	70.9	60.9	50.5	37.9	27.6	24.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	12.7	13.5	13.3	11.3	14.9	17.0	13.3	11.1	8.7	11.6	12.0	(35)
(36)			MCDBR	11.0	11.6	10.3	10.2	18.0	23.0	23.0	19.9	17.0	9.1	10.0	10.6	(36)
(37)			MCWBR	10.9	11.5	9.9	8.5	11.9	11.8	11.2	12.4	11.1	7.5	10.0	10.6	(37)
(38)		5% WB	MCD BR	11.1	11.7	10.2	10.1	16.9	21.5	21.6	18.0	14.9	8.7	10.3	10.9	(38)
(39)			MCWBR	11.1	11.7	10.0	8.9	11.8	13.0	12.3	12.7	11.0	7.7	10.4	11.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.203	0.253	0.299	0.324	0.341	0.351	0.376	0.357	0.315	0.262	0.194	N/A	(40)	
(41)		taud	1.881	2.023	2.059	2.085	2.161	2.400	2.367	2.402	2.470	2.191	1.910	N/A	(41)	
(42)		Ebn at Noon	36	173	229	259	268	270	258	250	235	171	35	0	(42)	
(43)		Edh at Noon	4	17	29	37	39	32	32	28	20	14	3	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	13	144	623	1240	1676	1571	1572	965	548	169	32	0	(44)	
(45)		RadStd	2	28	100	113	172	161	236	104	55	30	4	0	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+3.29	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+3.33	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page